

Per QA Q11 / Q23: every external data source touching our pipeline (training, fine-tuning, retrieval, evaluation) is declared below.

## 1. Datasets used

| Source                            | Release / version                                                           | Where used                                                      | Volume                                            |
|-----------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------|
| <b>EXACT2026 official dataset</b> | 2026-05-15 (drop-in replacement for 2026-05-09; see CHANGELOG_TYPE{1,2}.md) | Training (90% split), internal self-eval (10% holdout, seed 42) | 1,352 physics + 808 logic questions (411 records) |

No other datasets are used. Specifically, **no** external academic corpora, **no** web-crawled data, **no** Vietnamese physics textbooks, **no** prior EXACT 2025 data.

## 2. Synthetic data from closed-source models

**None.** No GPT, Claude, Gemini, or other closed-source model was used to generate any training, fine-tuning, retrieval, or evaluation data at any stage of this submission.

## 3. Crawled / scraped data

**None.**

## 4. Knowledge base / retrieval corpora

The deployed system retrieves only from the official EXACT2026 release above. No external knowledge base is built or queried at inference.

If a retrieval-augmented variant is shipped (currently planned but not yet enabled): the retrieval corpus consists of the *training-split* rows (90% of the 2026-05-15 release) used as few-shot exemplars. No data outside section 1 is added.

## 5. Models used at inference

| Role                                                          | Model                                                                         | License                                              | Params                         | Hosting                                                                                                                      |
|---------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Utility extractor / NL->FOL translator / explanation phrasing | <b>Qwen2.5-3B-Instruct</b><br>(qwen2.5:3b-instruct,use; per QA Q4)<br>Q4_K_M) | Qwen License<br>(permissive research use; per QA Q4) | 3.1B (nominal 3B-class, <= 8B) | Self-hosted on rented GPU via Ollama (OpenAI-compatible /v1/models and /v1/chat/completions; verified by team before deploy) |

**Only one LLM is loaded at inference at any moment** (per QA Q3 rule of thumb). If a sequential second model is added (e.g. a specialised NL->FOL translator for Type 1), it will be declared here in the final version of this document.

No closed-source LLM is called at inference. No third-party hosted inference API (Together AI, Fireworks, Groq, Replicate, HF Inference, etc.) is used.

## 6. Fine-tuning / training disclosure

| Artefact                                                                                                                | Base | Data                                                                                                           | Recipe                                                             | Used at inference?                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| models/exact_qwen25_7b-unsloth/qwen2.5-7b- <del>instruct</del> train_mixed_solve_qwen2.5-7b-unsloth, (kept as ablation) |      | (1,765/196 train/val), derived from EXACT2026 official release 2026-05-09 (no augmentation, no synthetic data) | Qwen2.5-72B-Instruct, alpha=32, 2 epochs, Unsloth on free Colab T4 | <b>No</b> – omitted from deployed endpoint for latency (ADR 0013); retained as a documented, reproducible ablation only |

If the production deploy is upgraded to use the SFT'd adapter as a sequential specialised extractor (option E7a in our improvement plan), this row will be moved to section 5 and re-declared.

## 7. Tools (per QA Q7 – do not count toward the parameter limit)

- **SymPy** – symbolic algebra / numerical solver for physics formulas
- **pint** – SI unit normalisation and conversion
- **Z3** – first-order-logic entailment / theorem proving for Type 1
- **scikit-learn** TF-IDF / Jaccard similarity (premise selector)
- **FastAPI** / **uvicorn** – HTTP serving

All tool invocations are reflected in the `cot/explanation` fields of the API response so that evaluators can verify the reasoning trace (per QA Q7 visibility requirement).

## 8. Contact

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